

Guangzhou (David) Cai

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EDUCATION

Carnegie Mellon University

Master of Science in Electrical & Computer Engineering.

Pittsburgh, PA

December 2027

University of Rochester

Bachelor of Science in Electrical & Computer Engineering.

Rochester, NY

May 2025

SKILLS

Languages: JavaScript, Python, C/C++, MATLAB, SQL

Web/Backend/Databases: HTML/CSS, Django, FastAPI, REST APIs, SQLAlchemy, Pydantic, SQLite, PostgreSQL, Uvicorn

Cloud/Infrastructure: Git, GitHub, Docker, AWS/GCP, Apache Spark, Kafka, MLOps

AI/ML: PyTorch, scikit-learn, Keras, TensorFlow, Transformers, CNNs, RNNs

Relevant Coursework: Web Application Development, Systems Toolchains for AI Engineers, Building Reliable Distributed Systems, Machine Learning, Foundations of Computer Systems, Full-Stack Engineering

EXPERIENCE

Software Engineer Intern

May 2024 – July 2024

ADASECO (Advanced Driver Assistance Systems Safety Company)

San Gabriel, CA

- Led daily stand-ups and sprint reviews for a team of 8, presenting YOLO road-detection updates and capturing action items to improve vehicle and pedestrian detection accuracy on Raspberry Pi.
- Facilitated code reviews with senior engineers, documented decisions, and wrote follow-up tickets to unblock deployment.
- Authored a 2-page analysis brief on Titan dataset quality and feature selection and shared results with the team.

PROJECTS

AI Internship and Research Application Assistant

May 2026 – Present

<https://github.com/DCAI12313/ai-application-assistant>

Independent AI and Backend Engineering Project

- Built a modular FastAPI backend for managing applications, job descriptions, candidate profiles, fit analyses, and generated reports.
- Implemented persistent CRUD workflows with SQLAlchemy and SQLite, including relational links between applications, jobs, profiles, analysis results, and saved fit reports.
- Developed resume ingestion from pasted text and PDF uploads using pypdf and rule-based extraction to identify technical skills and project experience.
- Created a synonym-aware job-matching engine that generates fit scores, matched and missing skills, strengths, gaps, and recommended actions through REST APIs.

AI Systems & Data Infrastructure

Aug. 2026 – Present

Python, PySpark, Apache Spark, PostgreSQL, SparkSQL, Google Cloud, Git

- Developing data and ML infrastructure workflows using Python, PySpark, Apache Spark, PostgreSQL, and SparkSQL, with emphasis on scalable data ingestion, transformation, and processing.
- Preparing and analyzing the NSL-KDD network intrusion dataset across 41 input features and multiple attack categories.
- Building cloud-oriented data pipelines that integrate relational storage with Spark DataFrames for distributed data preparation, cleaning, and feature engineering.
- Designing production AI system architecture incorporating LLM agents, tool orchestration, persistent memory, knowledge bases, and external API integration.

Web Application Development

Aug. 2026 – Present

Python, Django, JavaScript, HTML/CSS, HTTP, Git/GitHub

- Developed interactive web applications using HTML/CSS and JavaScript, implementing DOM manipulation and event-driven client-side behavior.
- Built Django-based server applications using HTTP request/response handling, URL routing, views, templates, static assets, and MVC separation of concerns.
- Developed a team-developed dynamic web application with persistent user data, iterative sprint demonstrations, and backend-focused software architecture.